

Project Design Template

One sheet, three gates, and four tests that say whether to start

1 What is actually missing

Question 34 wants a detour out of one project to make a connection to neighbouring methods concrete, and its note says the process of project designing should be standardized first. The research statement already names the shortage precisely: it is not a shortage of projects, it is a shortage of sharpened arguments connecting challenge, method, and contribution. Ideas arrive in free-form brainstorming and stay there. The sheet below is the shape they get poured into.

Two design decisions before the sheet, both taken from the Think page.

A list of heuristics is not the deliverable. General heuristics transfer badly, because knowing an operation and regulating when to use it are different capacities. So this is a procedure with stopping conditions rather than a list of moves, and the stopping conditions are the part that does the work.

Read the stage list as a menu, not a mandate. The useful move is to walk it once at the start of a project and write down which stages you are consciously not doing. A stage skipped on purpose costs nothing. A stage skipped without noticing is found in month five.

2 The three gates

These are the author's own, and they are prior to everything on the sheet. A project kicks start and materializes when all three are met, and not before.

1. **Experiment infrastructure implemented.** Not designed. Running.
2. **First evidence for the hypothesis in hand.** Something already observed that would be strange if the hypothesis were false.
3. **Algorithm sketch ready.** Concrete enough to hand over.

The third gate exists for a practical reason, and the reason is worth stating in the file so it does not get softened later. Students come from colleagues rather than from a group of one's own, and an idea that is not concrete enough will not attract them, nor will it be executable by them without the algorithm and the expected results handed over directly. The gate is not a standard of rigor. It is a condition for the work happening at all.

3 The sheet

Fifteen fields. Most are one line. The order matters, because each field constrains the next, and a field that cannot be written is the finding.

1. **The question, written so it could be wrong.** If no observation would settle it, it is a topic and not a question. Rewrite until it names something that could turn out either way.
2. **The two fields it sits between.** A question worth taking up usually sits where two literatures make different assumptions about the same object. Name both.
3. **The challenge, in plain language.** Before any terminology. If it cannot be said plainly it has not been understood yet.
4. **The terminology, refined until it searches.** Refine it until it matches the literature well enough to search and compare. This is also where the two fields in field 2 are checked for one term carried in two senses, which is itself a candidate project.
5. **Why it matters.** The scientific reason, not the funding reason.
6. **The design move.** The method, representation, or design decision used in response to the challenge. One sentence. If it takes three, there are probably two projects here.
7. **The borrowed method, and which of three uses.** There are three and they are not interchangeable. Integrate it as a device that makes the method work better. Take the algorithmic pattern and apply the same idea to your own model. Or take how they designed the experiments that showed their approach was better. Say which one, because the three cost very different amounts of work.
8. **Contribution, and its evidence.** Stated together or not at all.
9. **The invariant.** Not only what this project contributes, but what it helps establish that later work stands on: a problem class, a representational commitment, a methodological style, or an application domain in which later work stays credible.
10. **Requirements, each with its verification.** Write what the experiment must be able to show before writing the code that will show it. Every requirement on the way down has a verification on the way up. If you cannot say how a requirement will be verified, it is not a requirement, it is a hope.
11. **Decision gates.** Explicit points where you decide whether to continue, with the criteria fixed in advance. Research projects usually have no gates at all, which is why they end by exhaustion rather than by decision.
12. **Technical risk, tracked.** Likelihood times consequence, revisited, with the mitigation named beside each one.
13. **Interfaces.** Most integration failures sit at boundaries between components built by different people to different assumptions. True of spacecraft, true of a simulator and a policy learned somewhere else. List the boundaries and who owns each side.
14. **Venue slice.** Which claim goes to which venue, and the deadline.
15. **Unsupported claims.** The ones that need more reading or more data, listed rather than smoothed over. This field is never empty, and an empty one means the sheet was filled to look finished.

4 The four tests

Fields 1, 5, and the stopping rule below are the regulation layer. They are what people lack, as distinct from the operations they govern, and they are written as tests rather than as prompts for that reason.

1. **Could it be wrong?** Field 1. Name the observation that would settle it.

2. **Would the answer change something?** Name what would be done differently under each answer. A question qualifies the way a number does, by changing a decision.
3. **Is it reachable?** A question is reachable when an experiment you could run this year would move it. Otherwise it is a five year question and belongs on the vista rather than in an agenda.
4. **Has the space stopped producing candidates?** Stop when a full pass over both literatures adds nothing new. What remains then is choosing rather than searching.

5 Venues, as slices rather than as targets

One project usually carries more than one claim, and the claims are judged by different people. Slicing them is a design act and it belongs on the sheet rather than at submission time. The pattern already in use: the system paper to the human-computer interaction venue, the algorithmic contribution to the multiagent venue, the learning outcome to the education venue, and the cognitive claim to the cognitive science venue. Each gets the claim its reviewers can actually judge, and the earliest deadline is the one that has to clear all three gates at once.

6 Validation by convergence

Where the object being measured has no agreed definition, no single method establishes it. Validate by convergence across complementary methods of elicitation, each with a different blind spot. Agreement among independent methods confers confidence in an inferred object, and disagreement localizes where the inference is unreliable. The object is fixed not by one measurement but by the convergence of several. Five methods are already named for this: self-report, structured pairwise choices, process tracing through information boards, feature perturbation, and think-aloud protocols. Field 10 is where this lands. A requirement whose verification is one measurement of an undefined object has not been verified.

7 What would kill it

The most useful thing a design sheet can carry is the condition under which the project is not worth doing, written before the work starts. The form that condition takes is a comparison against the simpler alternatives, stated as a rule rather than as a hope.

The added component is supported only if it resolves a documented failure more reliably than every simpler alternative under the stated assumptions. If a simpler alternative supplies the same benefit, the evidence supports that architecture instead. If no alternative succeeds either, the result identifies an unresolved problem but does not by itself show that the added component is the missing piece.

The alternatives have to be listed by name in field 11 and each one has to be a gate. A comparison set assembled after the results are in is not a comparison set.

8 The macros

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% One sheet. #1 the project's short name.
\newcommand{\pdsheet}[1]{\subsection*{#1}}

% The fields, kept as macros so a missing one is greppable.
\newcommand{\pdfield}[2]{\textbf{#1.} #2}

% A requirement and the verification that closes it. Never one without the other.
\newcommand{\pdreq}[2]{#1, verified by #2}

% A gate. #1 the decision point, #2 the criterion fixed in advance.
\newcommand{\pdgate}[2]{#1, continue only if #2}

% A risk. #1 the event, #2 likelihood times consequence, #3 the mitigation.
\newcommand{\pdrisk}[3]{#1 (#2), mitigated by #3}

% A stage from the systems engineering list that is being skipped on purpose.
\newcommand{\pdskip}[2]{#1, skipped because #2}

% A claim and the venue that can judge it.
\newcommand{\pdvenue}[3]{#1 to #2, deadline #3}
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9 Filling it from a brainstorm

Question 34's note says to work from previous free-form chat histories. The order that turns that material into the sheet is seven steps, and it is already written in the gap plan.

1. Collect the informal intuition from notes, drafts, and meetings.
2. Identify the underlying challenge in plain language.
3. Refine the terminology until it matches the literature well enough to search and compare.
4. State the scientific reason the challenge matters.
5. Isolate the actual method, representation, or design move used in response.
6. State the concrete contribution and its evidence.
7. List the unsupported or weakly supported claims that require more reading or data.

Steps 1 through 7 fill fields 1 through 8 and field 15. Fields 9 through 14 are not in the brainstorm and never will be, because they are about execution rather than about the idea. Those are the ones that take the high energy hours.

One rule on the source material. The brainstorm is in your words, which is the reason it is worth keeping. Carry the sentences across rather than rewriting them. A sharpened argument built out of restated sentences is a different argument, and the restatement is where the sharpening stops.

10 Sources

- Fields 10 through 13, and the menu rule in section 1, are the five parts of the NASA Systems Engineering Handbook that transfer to a lab, as already written on the Project Management page of the website.
- The regulation argument, the four tests, and the stopping rule are from the Think page and from the problem finding note behind it.
- The seven step procedure in section 8, the invariant question in field 9, and the statement of what is missing in section 1 are from the gap plan section of the research statement.
- The three gates, the resource condition behind the third, the three uses of a borrowed method in field 7, and the venue slicing pattern are the author's own, from the project notes for the teaching game and for the world model failure work.
- The convergence argument and the five elicitation methods are from the validation section of the same project notes.
- The rule in section 7 is the decision rule already written for the agentic defender comparison, generalized here from causal integration to any added component.